

Laura Werthmann Dozal

• Email: lwerthmann@arizona.edu • <https://github.com/lwdozal> • LinkedIn: [./laurawerthmann/](https://www.linkedin.com/in/laurawerthmann/)

EDUCATION

PhD in Information Science; Minor in Statistics and Data Science; *Outstanding PhD Student Award 2026*

College of Information, University of Arizona, Tucson, Arizona - May 2026

Master of Arts in Communication, Culture, and Technology; *Georgetown University, Washington, D.C. 2016*

Bachelor of Arts in English Literature; Minor in Communication; *St. Edward's University, Austin, Texas 2012*

Bilingual: Fluent in English and Spanish

TECHNICAL SKILLS

Languages: Python, R and Quarto, SQL, unix, Javascript

Machine Learning and Artificial Intelligence: PyTorch, Keras, Transformers, LLMs (Llama 3, GPT-4), M-LLMs (Llama4 Scout, BLIP-2, phi4) Computer Vision (OpenCV, ViT, CNNs), RAG, GraphML, Graph Neural Networks (GAT, VAEs), applied machine learning, machine learning infrastructure, Hugging Face

MLOps/LLMOps & Engineering: Docker, DuckDB, Kubernetes (implied by node management), Langchain, Human-in-the-Loop implementations, AWS Cloud Infrastructure, Git/GitHub, CI/CD pipelines, Linux/HPC Administration, MLFlow, ETL data management.

Data & Viz: Pandas, NumPy, tidyverse, ggplot2, Scikit-learn, igraph, VSCode, Leaflet.js, Tableau, PowerBI

PROFESSIONAL EXPERIENCE

Teaching (Instructor on Record) | *University of Arizona College of Information - Tucson, AZ, 08/2019 – Present*

- ISTA 116: Statistical Foundations for the Information Age, *Spring/Fall Semesters 2024-Present*;
- ISTA 130 – Computational Thinking and Doing (Introduction to Python), *Summers 2021 - Present*;
- ESOC 300 – Digital Storytelling and Culture, *Summer 2020*
- **Graduate Teaching Assistant:** INFO 523: Data Mining and Discovery; INFO 523: Data Analysis and Visualization, Tucson, AZ, Fall 2023 – Spring 2024.

ML/AI Research (Fellow) | *Jetstream2 (NSF Funded), University of Indiana; Remote 1/13/2025 – 1/06/2026*

- *Compute and Processing Pipeline and Deployment:* Architected an end-to-end LLMops/MLOps pipeline on high performance computing clusters (90 GPU-enabled nodes, NVIDIA A100s) for multimodal classification. Built reproducibility workflows to standardize data collection and computer vision model training.
- Lead Open Science project that holds a version control codebase (GitHub, docker, etc) for data collection, computer vision classification models, semantic clustering, and graph database development. Implemented a comparative analysis of CNNs, vision transformers (ViTs), multimodal (vision-language), and generative models large foundation models and SOTA AI/ML frameworks and toolkits via PyTorch, Tensorflow, Langchain, huggingface, and OpenCV.
- Deployed and fine-tuned open-source multimodal generative AI models (LLAMA 3 and 4, phi4, Qwen2.5 VL, BLIP-2) for image labeling and captions. Generative model evaluation of text descriptions and image labels, using accuracy metrics and semantic similarity (BERTScorer and SBERT similarity). Evaluated AI models by running benchmark experiments, analyzing data and implementing human-agent-interaction and human in the loop methods. Applied graphical analysis on image and text embeddings using Variational Autoencoders (VAE) within a Graphical Attention Network (GAT) model to compare classifications of the collected annotated image data to other benchmark graph datasets, Cora, flicker. Evaluated all VAE embeddings using visualization on the image data for unsupervised learning support.

Data Science / Machine Learning Research Intern | *Laboratory for Analytic Sciences - Summer Conference of Applied Data Science; North Carolina State University, Raleigh, North Carolina, 06/2024 – 07/2024*

- *Compute and Processing Used:* Unlimited AWS Cloud Infrastructure and storage services. Unlimited access to g1-g4 HPC instances. 3+ TB of storage. **Awarded \$11,500 Stipend.**
- Project lead for computer vision research team using video object tracking models: zero-shot transformers (OWL-ViT, GPT-4o, Phi3) on noisy real-world video surveillance data. The goal was to explore improvements for tracking object obfuscation and detecting unknown/unconventional objects in the noisy video data.

PROFESSIONAL EXPERIENCE (Continued)

- Engineered a Retrieval-Augmented Generation (RAG) system using LangChain architecture to evaluate text segmentation methods for text summarization, implementing a vector embedding database for high-efficiency retrieval. Used RAGAs evaluation framework on 2024 TREC data-MSMarco docs and QA.

Data Science Researcher (Fellow) | UofA Data Science Institute, Roots for Resilience, 08/2023 – 01/2024

- Compute and Processing Resources Awarded: 3 TB of data storage; 20,000 compute units/year; Ability to run 4 concurrent jobs with instance allocation. **\$6,000 Grant**
- Data Management**: Created and maintained a Data Management Plan for social media data project, using Open Science Framework in collaboration with CyVerse (cyberinfrastructure/DataOps), GitHub, and Docker.
- ETL infrastructure for interactive web interface. Used SQL queries and database management to connect MySQL database to leaflet map built in JavaScript using Python library, flask API, Interactive Leaflet map creation for social movement group in Mexico Ellastienennombre.org. Created search feature to retrieve map data points of victims in cd, Juarez.

Data Science Intern | Drexel University / University of Rochester Remote, 2023 LEADING Program, 06/2023 – 01/2024

- Led the creation of an Open Source Data Pipeline to collect and analyze bibliometric data by University of Rochester Researchers to identify their publication behavior. Collected, cleaned, and analyzed 500K+ dataset metadata using APIs from various repositories. Used trend and network analysis methods including regression (decision trees), time series analysis, structural analysis, and prediction analysis.

Data Science Grad Assistant | University Analytics & Institutional Research, Tucson, AZ 08/2021 – 08/2023

- NLP Sentiment Analysis**: Analyze 4M+ student course survey responses using Natural language processing to identify sentiment, word frequencies and word synonyms for curriculum in various departments on campus. Increase faculty participation in curriculum review across schools on campus.
- Machine Learning** - Spearheaded the Women in Science and Engineering Retention report project which uses survival and churn analysis. These analyses review churn or dropout rates of 37K+ students over ten years. The analysis incorporates machine learning applications including decision tree and random forest models along with other time-based statistical methods. I identified student financial, cultural, academic, and demographic characteristics that either debilitate or foster success towards graduating college.
- Dashboard Creation** - Using automated and qualitative approaches, I developed End-to-end application of tuition and headcount dashboard visualizations and analysis using R and SQL. Created aggregated data tables from an internal Oracle data warehouse using descriptive statistics and inflation information of campus-wide Net Tuition Revenue (NTR). The dashboard is currently used by decision makers on campus.

Data Science Ambassador | University Arizona Data Science Institute, Tucson 08/2021 – 08/2022

- Data Carpentries Instructor as of 2/14/2022 -Live coding instructor and helper for Data Analysis and Visualization with R for Social Scientists, Introduction to Python, Open Refine, and other data science skills.
- Project Manager: data science community support and resources for the CSS certificate and iSchool. Created and Organized Projects in Progress. Fostered a Data Science community environment for researchers on campus to collaborate and share projects ideas and progress. Helped consult 3 - 5 projects over the span of two semesters (over 20 researchers) using data science methods monthly.
- Helped organize Stanford-based Women in Data Science satellite conference, organized and moderated the Computational Social Science panel. The April 2022 event had around 400 participants on/offline.

SELECT PUBLICATIONS

- The Library Quarterly** Forthcoming, 2026. "When VR Was New": Looking Back at Undergraduate Students' Initial Encounters with Virtual Reality in an Academic Library.
- Jetstream2 AI Workshop Project Manager** - February 2026. doi:10.5281/zenodo.18407373 "AI Pipeline for Synthetic Data Generation, Generative M-LLM Comparison, and Network Building"
- Association for Computational Linguistics (ACL)** - July 2024. Conference workshop presentation. "Visual and Textual Narrative Analysis of the Online Anti-Femicide Movement in Mexico: A Narrative Network Approach".

- **LREC Conference. Paper Publication May - 2022, Extracting Space Situational Awareness Events from News Text**